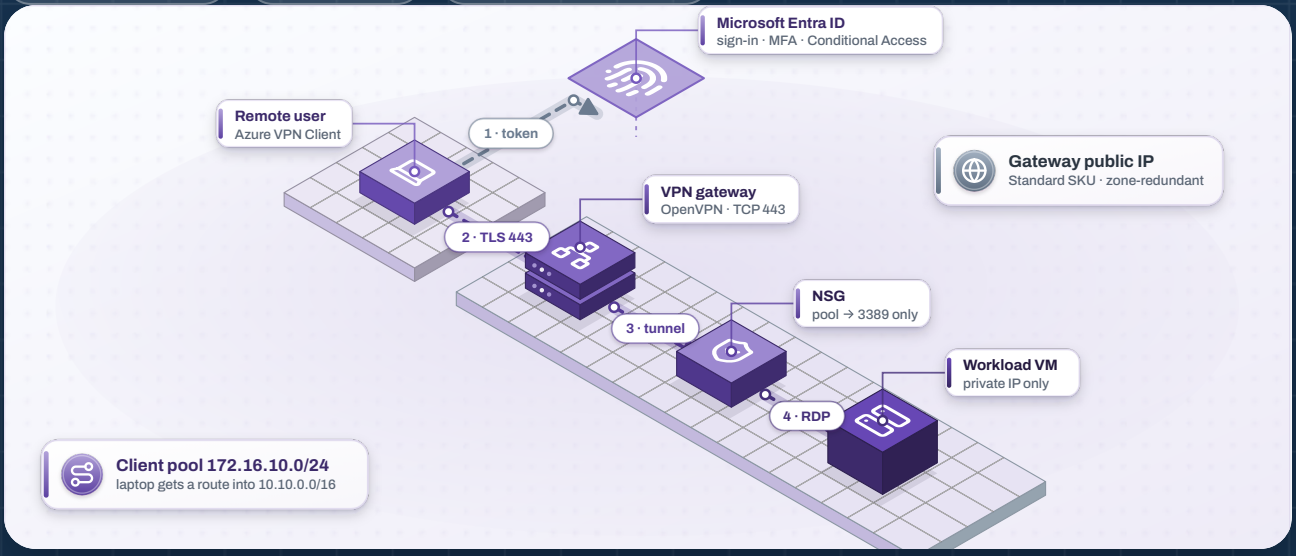
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-AZ-004

Azure Point-to-Site VPN with Microsoft Entra ID Authentication

Give remote users an MFA-protected, OpenVPN tunnel into a private Azure network, built four ways (portal, PowerShell, Azure CLI, certificate authentication) plus the end-user connection procedure.

Microsoft Azure | Networking | Identity & Zero Trust



DOCUMENT ID	SOP-AZ-004
VERSION	2.0
EFFECTIVE	October 2026
AUDIENCE	Cloud and network administrators, identity administrators, help desk
AUTHOR	Richmond Dieujuste, Principal Consultant

Rich Dieu IT Solutions LLC · richdieu.com

Sanitized reference edition — client identifiers removed

Contents

1. Concepts You Need First 3

1.1 Four ways to reach a private Azure network 4

1.2 The VPN gateway, GatewaySubnet and SKUs 4

1.3 Tunnel protocols and authentication types 5

1.4 Audience, admin consent and the Azure VPN Client app 6

1.5 Address pools and CIDR planning 7

1.6 Choosing a method 9

2. Access, Prerequisites and Planning 10

2.1 Level of access required 10

2.2 Prerequisites checklist 11

2.3 Decisions to make up front 12

3. Method A · Building It in the Azure Portal 13

4. Methods B, C and D · PowerShell, Azure CLI and Certificates 18

4.1 Method B · Az PowerShell base template 18

4.2 Method C · Azure CLI 20

4.3 Method D · Certificate authentication (alternative) 20

5. Method E · End-User Procedure, Verification and Monitoring 22

5.1 Connecting as an end user 22

5.2 Verification checklist 23

5.3 Sessions and logs 23

5.4 Control alignment (NIST SP 800-53 Rev. 5) 24

6. Rollback, Cost and Troubleshooting 25

6.1 Rollback and cost 25

6.2 Troubleshooting 25

7. Summary, Glossary and Review 27

7.1 Chapter summary 27

7.2 Glossary 27

LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Explain when to choose P2S VPN over Site-to-Site VPN, ExpressRoute or Azure Bastion.
- Plan non-overlapping VNet, GatewaySubnet and client-pool ranges and pick a gateway SKU.
- Compare OpenVPN, IKEv2 and SSTP, and Entra ID, certificate and RADIUS authentication.
- Build the gateway in the portal, with an Az PowerShell template and with the Azure CLI.
- Choose between the Microsoft-registered and legacy VPN audiences and handle admin consent.
- Enforce MFA with Conditional Access, verify the connection and troubleshoot failures.

TIME	DIFFICULTY	MINIMUM ACCESS	RISK
≈ 30 min active + 30–45 min gateway deploy	Intermediate	Network Contributor (RG) + Cloud Application Administrator (Entra)	Low (new resources; gateway has ongoing cost)

CHAPTER 1

Concepts You Need First

A virtual machine in Azure that has **no public IP address** is unreachable from the internet. That is exactly what you want for security, but people still need to reach it. A **Point-to-Site (P2S) VPN** builds an encrypted tunnel from *one device* (a laptop) to a **VPN gateway** inside your Azure virtual network (VNet). Once the tunnel is up, the laptop behaves as if it were plugged into the VNet: it receives a private IP address and can open Remote Desktop (RDP), SSH or any other protocol to the VM's *private* address.

KEY CONCEPT · A STAFFED ENTRANCE, NOT AN OPEN DOOR

Think of the VNet as an office building with no street-level doors. The VPN gateway is the single staffed entrance. Each visitor shows a badge (their Microsoft Entra ID account), passes a second check (MFA), and only then receives a visitor sticker (an IP from the client pool) that lets them walk the corridors. Nobody can reach a desk (a VM) from the street, because no desk has a street door.

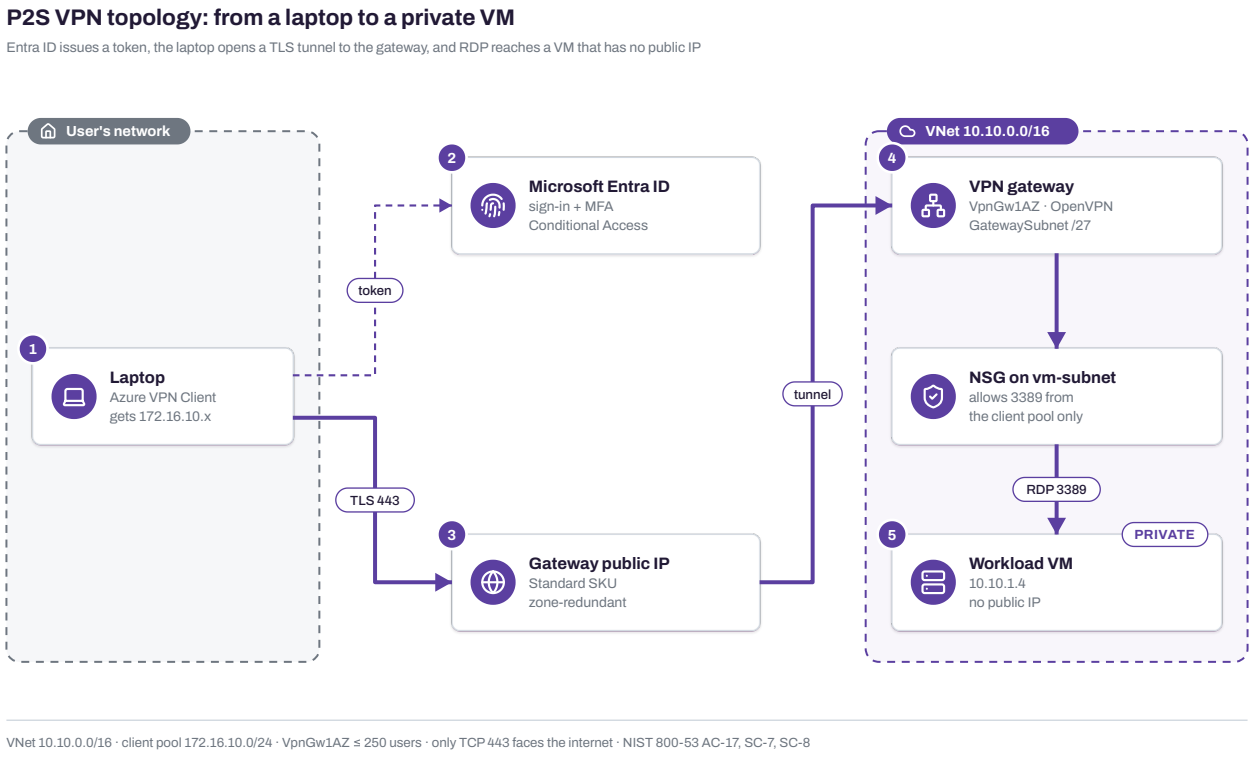


Figure 1. The VM never needs a public IP: the gateway is the only internet-facing component, and every connection is tied to an Entra sign-in.

1.1 Four ways to reach a private Azure network

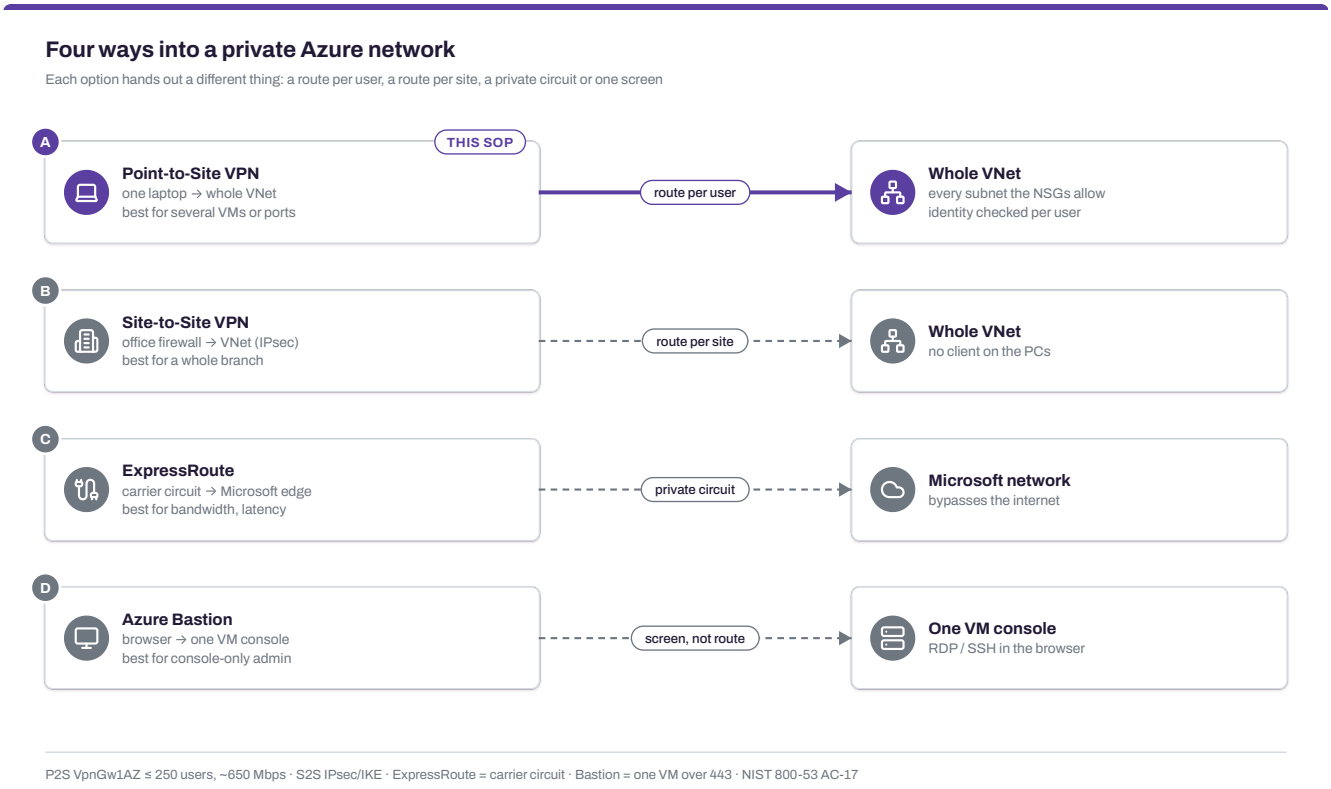


Figure 2. The more a door opens, the stronger the identity check it needs: P2S hands each user a route, so it is protected by MFA and Conditional Access.

Site-to-Site VPN links a whole office firewall to a VNet over IPsec, with no client on the PCs. **ExpressRoute** is a private carrier circuit into Microsoft's network for high bandwidth and predictable latency. **Azure Bastion** gives browser-based RDP/SSH to one VM at a time, a *screen* rather than a route. **P2S** gives each user's laptop a *route* into the whole VNet (subject to NSG rules), which is more flexible and therefore needs stronger identity controls. Rule of thumb: individuals needing several ports or VMs → P2S; console-only access → Bastion; a whole site → Site-to-Site; private, high-bandwidth links → ExpressRoute.

1.2 The VPN gateway, GatewaySubnet and SKUs

A **virtual network gateway** of type *VPN* is a managed service: Azure runs hidden VM instances behind it, patches them and fails them over. Those instances need addresses, so Azure requires a dedicated subnet named exactly `GatewaySubnet` (/27 or larger recommended). Nothing else may live there, and the VPN type must be **route-based** for P2S.

The **SKU** sets throughput, the maximum number of concurrent P2S connections and the feature set. "AZ" SKUs spread instances across availability zones, so one datacenter outage does not take the gateway down. Microsoft is consolidating gateways onto AZ SKUs with

Standard public IPs (Basic public IPs are retired), so build new gateways on an AZ SKU and confirm current limits on the official *VPN Gateway SKUs* page.

SKU	Max P2S connections	Approx. throughput	Notes
Basic	n/a for this SOP	~100 Mbps	Legacy; no OpenVPN, no Entra ID authentication.
VpnGw1AZ	250	~650 Mbps	Used in the reference build; small and medium teams.
VpnGw2AZ / 3AZ	500 / 1,000	~1 / ~1.25 Gbps	More users or heavier transfers.
VpnGw4AZ / 5AZ	5,000 / 10,000	~5 / ~10 Gbps	Generation 2 only; enterprise scale.

1.3 Tunnel protocols and authentication types

Protocol	Transport	Strengths and limits
OpenVPN	TLS over TCP 443	Looks like HTTPS, so it passes almost any hotel or corporate firewall. **The only protocol that supports Entra ID authentication.** Uses the Azure VPN Client.
IKEv2	IPsec, UDP 500/4500	Native in Windows, macOS and iOS; fast reconnects. Often blocked on restrictive networks. Certificate or RADIUS auth only.
SSTP	TLS over TCP 443	Windows-only, Microsoft proprietary, lower connection limits. Legacy.

Encryption makes traffic private; **authentication** proves who is on the other end. A gateway can accept more than one type:

- **Microsoft Entra ID**: the Azure VPN Client signs the user in and presents an access token; MFA and Conditional Access apply automatically, and disabling the account blocks new connections.
- **Certificate**: each device holds a client certificate chained to a root whose public key is uploaded to the gateway; revocation is by thumbprint (Method D).
- **RADIUS**: the gateway forwards credentials to a RADIUS server such as Windows NPS, which checks Active Directory; MFA via the NPS extension.

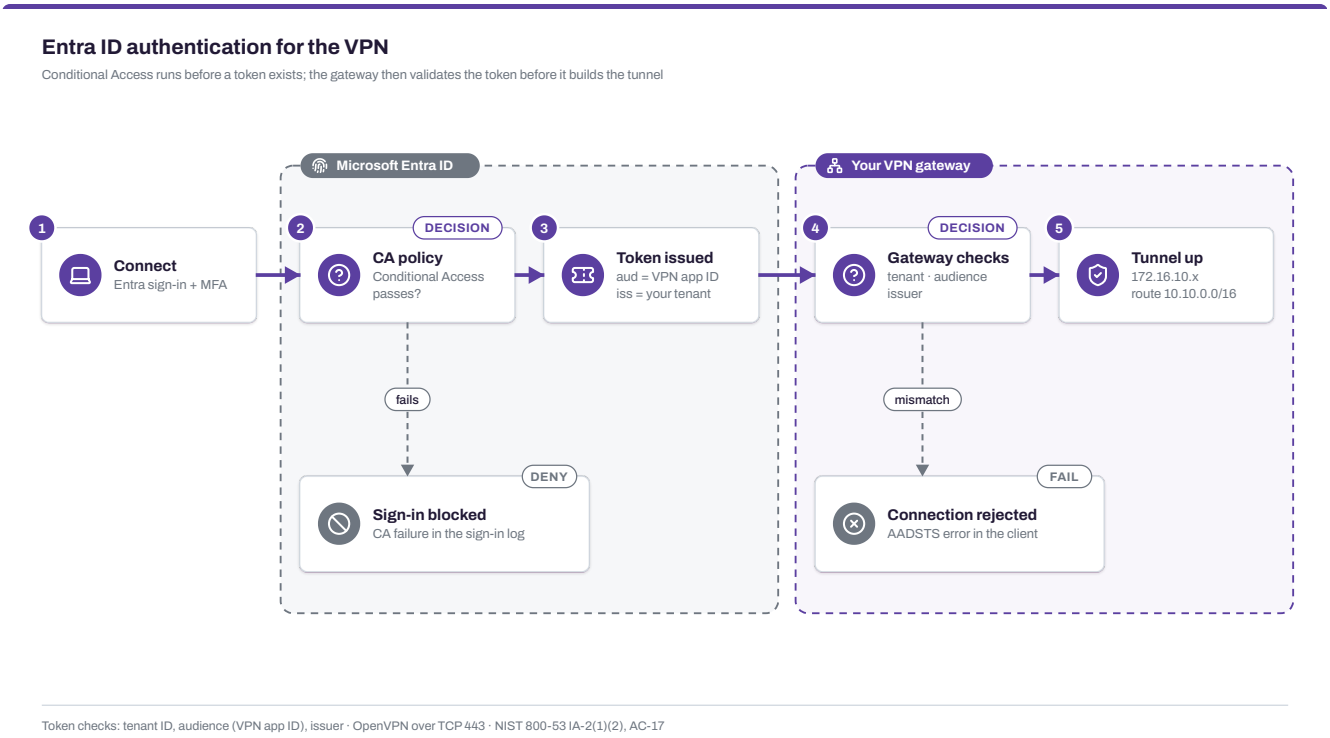


Figure 3. A blocked sign-in and a rejected connection fail at different points, so check the Entra sign-in log first and the client error second.

1.4 Audience, admin consent and the Azure VPN Client app

An Entra access token is always issued *for* a specific application, identified by its application (client) ID. The gateway's `Audience` field tells it which ID to expect. Two Microsoft IDs exist:

Audience value	What it is	Consent and client requirements
`c632b3df-fb67-4d84-bdcf-b95ad541b5c8` **Microsoft-registered Azure VPN Client**	Current, recommended ID. Microsoft registers it, so nothing has to be created or authorized in your tenant first.	No separate admin-consent step. Needs a recent Azure VPN Client and a freshly downloaded profile; check Microsoft's supported-client list.
`41b23e61-6c1e-4545-b367-cd054e0ed4b4` **Manually registered "Azure VPN"**	Original ID used by older configurations, including the reference build.	Requires one-time **tenant-wide admin consent** (Chapter 3, Step 5), after which an *Azure VPN* enterprise app appears. Works with older clients. The consent covers only delegated sign-in and basic-profile permissions, not mail, files or admin rights.

A **custom audience** (your own app registration or custom scopes) can restrict connections to assigned users, but most organizations get the same effect more simply with `Assignment required` on the enterprise app plus Conditional Access.

> **WATCH OUT · DO NOT MIX AUDIENCES**
>
> The gateway's audience and the one inside users' profiles must match. Moving a gateway from `41b23e61-…` to `c632b3df-…` means re-downloading and redistributing the profile and retargeting any Conditional Access policy at the new enterprise app. Treat it as a change, not a tweak.

Conditional Access and MFA. Because Entra ID performs the sign-in, every tenant identity control applies to the VPN with no extra infrastructure. A Conditional Access (CA) policy targeting the VPN enterprise app can require MFA, a compliant device, low sign-in risk or approved countries. Disabling a user or revoking their sessions stops new connections immediately.

1.5 Address pools and CIDR planning

Three IP ranges must be planned together. CIDR notation (`10.10.0.0/16`) gives a start address and the number of fixed leading bits: /16 = 65,536 addresses, /24 = 256, /27 = 32. Azure reserves five addresses per subnet.

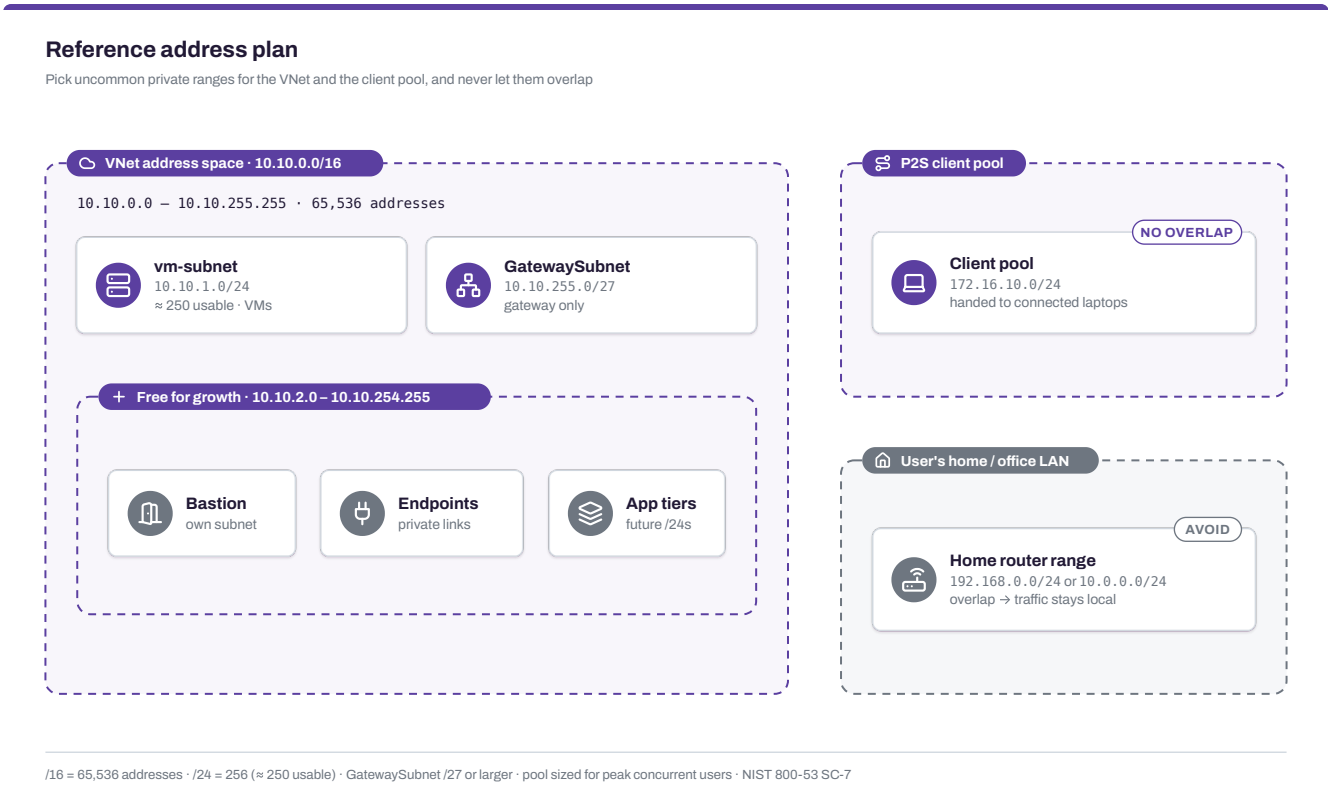


Figure 4. Overlapping ranges fail silently, because the laptop keeps traffic for its own LAN local instead of sending it through the tunnel.

WORKED EXAMPLE · WHY OVERLAP BREAKS THINGS

A user's home router hands out `10.10.1.0/24` and the Azure VM is `10.10.1.4`. Windows sees that address as local and sends RDP to the living room instead of the tunnel. The client shows *Connected*, yet RDP times out. Choose uncommon ranges, and size the pool for peak concurrent users (a /24 supports about 250).

1.6 Choosing a method

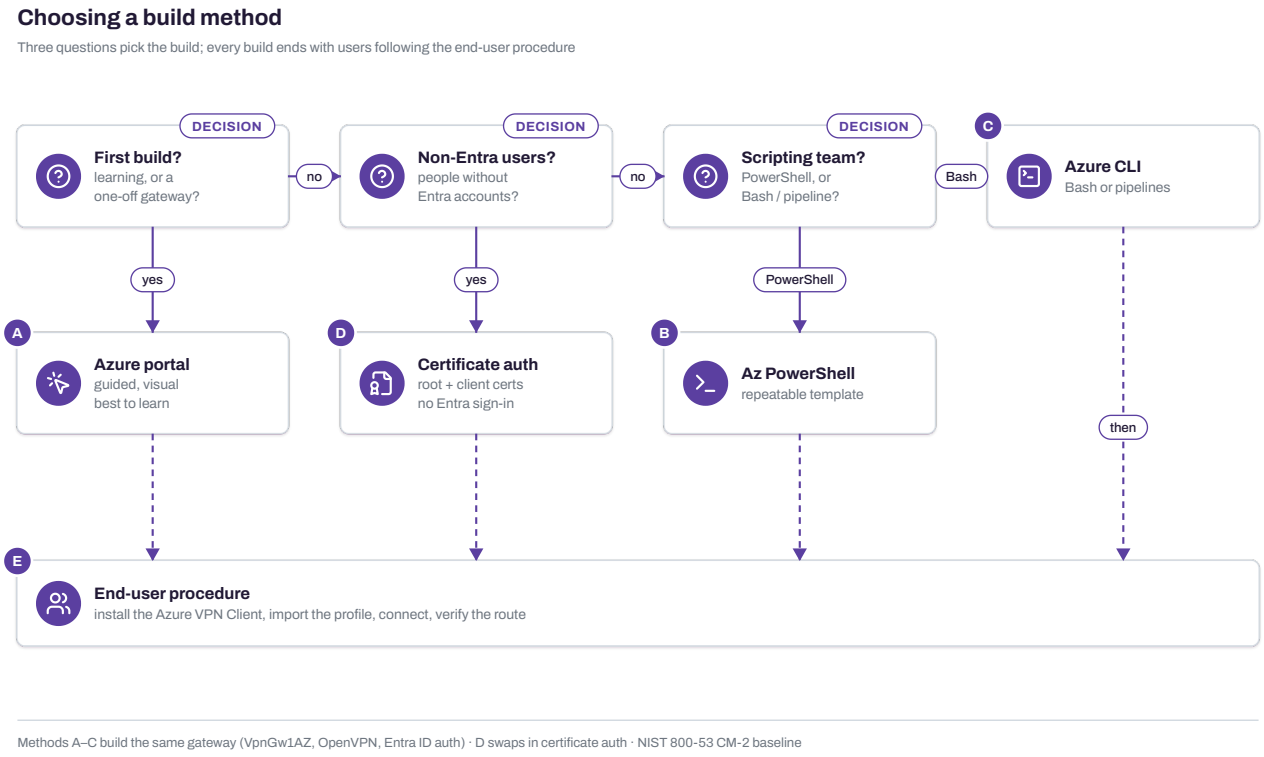


Figure 5. All build methods produce the same gateway, so pick the one your team can repeat; users then follow the same end-user procedure (Method E).

A · Azure portal

Wizards for VNet, gateway and P2S settings, shown with screenshots.

BEST FOR: FIRST BUILD, LEARNING

B · Az PowerShell

One parameterized script builds it all, downloads the profile, writes evidence.

BEST FOR: REPEATABLE, AUDITED BUILDS

C · Azure CLI

The same build in Bash for Cloud Shell and pipelines.

BEST FOR: LINUX ADMINS, CI/CD

D · Certificate auth

Root and client certificates instead of Entra ID.

BEST FOR: NON-ENTRA USERS, BREAK-GLASS

E · End-user procedure

Install the client, import, connect, RDP.

BEST FOR: EVERY USER, HELP DESK

CHAPTER 2

Access, Prerequisites and Planning

2.1 Level of access required

This build touches two permission systems. **Azure RBAC** governs network resources and is granted at a scope such as a resource group. **Microsoft Entra roles** govern tenant-wide identity objects: enterprise apps, consent grants, Conditional Access. Owner on a subscription cannot grant admin consent, and a Global Administrator has no Azure resource rights by default.

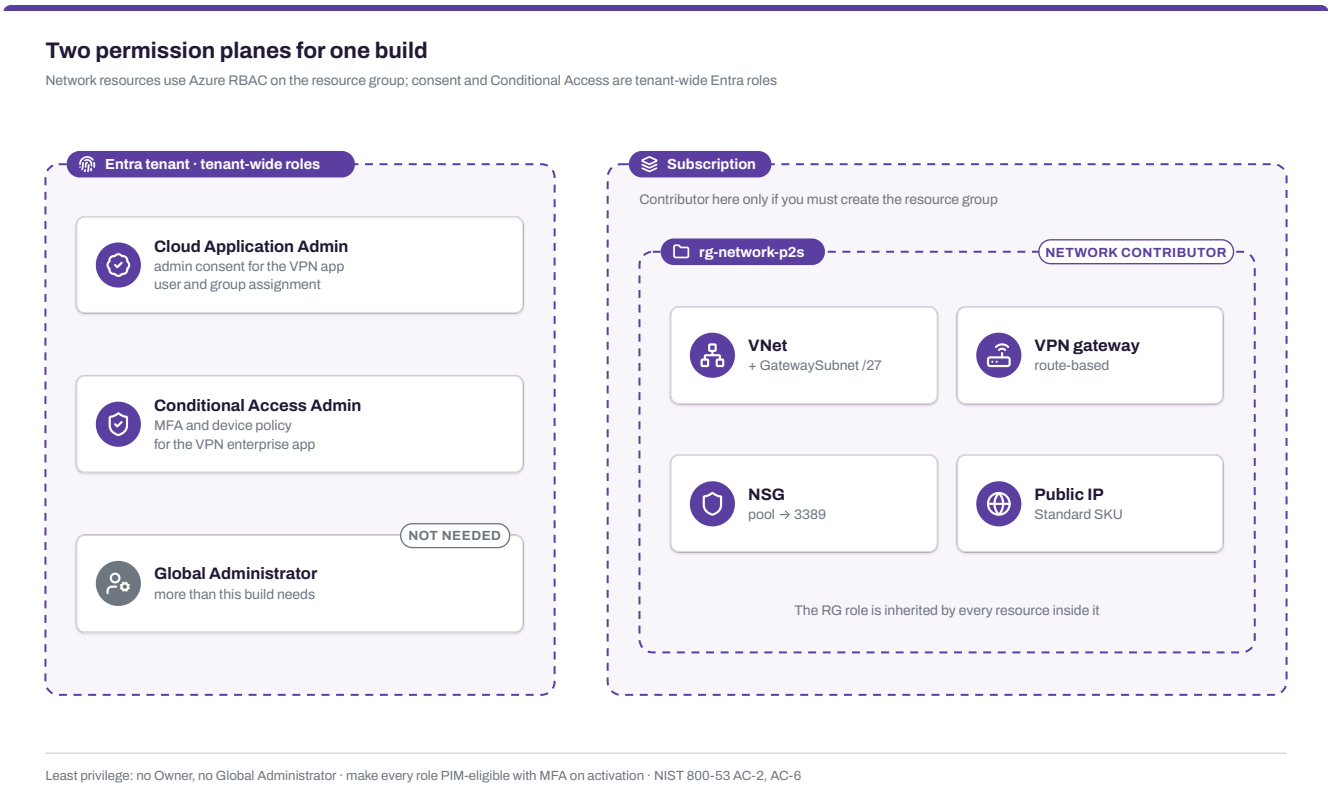


Figure 6. Owner on a subscription cannot grant admin consent, and a Global Administrator has no network rights by default, so plan both role planes before the change window.

Task	Least-privilege role	Scope and notes
Create VNet, subnets, NSG, public IP, gateway; configure P2S; download profile	Network Contributor (Azure RBAC)	Resource group `rg-network-p2s` (Contributor on the subscription only if you must create the RG).

Grant admin consent (legacy audience only)	Cloud Application Administrator or Application Administrator (Entra)	Tenant. Enough for delegated permissions; Global Administrator is more than needed.
Require assignment, assign the VPN group	Cloud Application Administrator or app owner	The VPN enterprise application.
Create the Conditional Access policy	Conditional Access Administrator	Tenant (Security Administrator also works).
Gateway diagnostics to Log Analytics	Monitoring Contributor + Log Analytics Contributor	Gateway and workspace.
View sessions / disconnect a user	Reader / Network Contributor	Gateway resource; useful for help desk escalation.
End user connects and uses RDP	No Azure role; assigned to the VPN app and passing CA	Inside the VM: *Remote Desktop Users*, or *Virtual Machine User Login* for Entra-joined VMs.

GOOD PRACTICE · JUST-IN-TIME PRIVILEGE

Make Network Contributor and the Entra roles *eligible* in Privileged Identity Management (PIM). Administrators activate them for a few hours with MFA and a change number, and the activation history documents who held which privilege during the build.

2.2 Prerequisites checklist

- [ ] An approved change record (e.g. `CHG0012345`), applied as a `ChangeRef` tag on every resource.
- [ ] A target region that supports AZ gateway SKUs, in the same region as the workloads.
- [ ] An address plan agreed with networking: VNet, workload subnet, GatewaySubnet, client pool; no overlap with existing, peered or common home ranges.
- [ ] Your Entra **tenant ID** (Microsoft Entra ID › Overview) and a chosen audience (Microsoft-registered for new builds).
- [ ] Users enrolled in MFA, and a security group of permitted users, e.g. `SG-VPN-P2S-Users`.
- [ ] Budget approval: a VPN gateway bills every hour it exists, whether or not anyone is connected.

2.3 Decisions to make up front

Decision	Guidance
SKU	VpnGw1AZ up to ~250 concurrent users; resize upward later within the same generation.
Tunnel and auth	OpenVPN + Entra ID for managed users; add certificates only for users who cannot use Entra ID.
Split vs forced tunnel	Default is **split tunnel**: only VNet routes use the VPN. Forced tunneling needs extra configuration and capacity.
DNS	Clients use the VNet's DNS servers. To resolve Azure Private DNS zones, deploy Azure DNS Private Resolver or a forwarder in the VNet.

CHAPTER 3

Method A · Building It in the Azure Portal

Six stages. Steps 1–2 take minutes, Step 3 takes 30–45 minutes of waiting, Steps 4–6 a few minutes each. Sign in to **portal.azure.com** with MFA and activate your PIM roles first.

STEP 1 Create the virtual network

1. Search **Virtual networks** › **+ Create**. **Basics**: subscription, resource group `rg-network-p2s`, name `vnet-hub-p2s`, region.
2. **Security**: leave Bastion, Firewall and DDoS Network Protection off unless your design needs them (each adds cost).
3. **IP addresses**: set the space to `10.10.0.0/16`; edit the *default* subnet to `vm-subnet`, `10.10.1.0/24`.
4. **Tags**: `ChangeRef`, `Owner`, `Environment`. Then **Review + create**, check the values, **Create**.

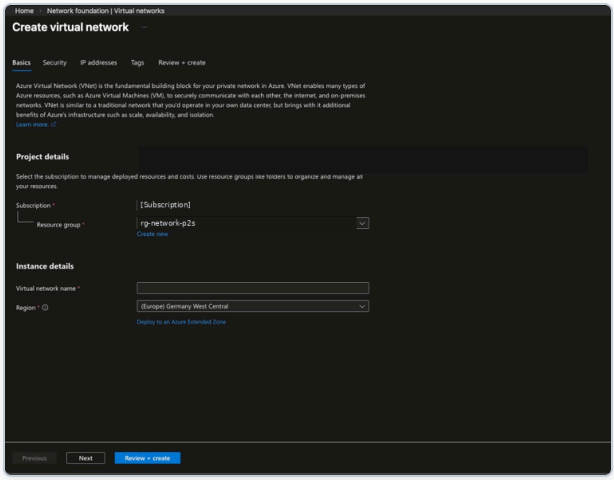


Figure 7. Basics tab. The resource group and region decide where the gateway must also live.

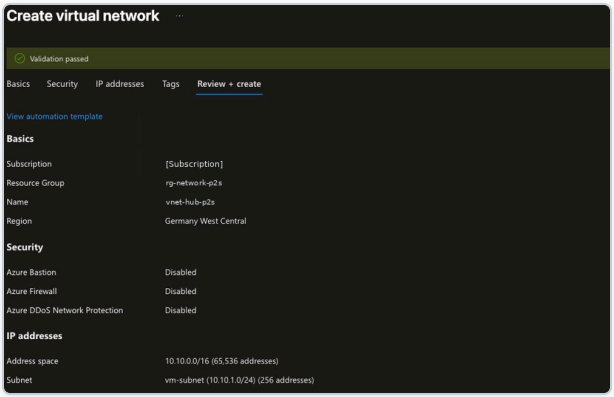


Figure 8. Review + create: address space /16 and workload subnet /24.

STEP 2 Add the GatewaySubnet

1. In the VNet open **Settings › Subnets › + Subnet**, set **Subnet purpose** to **Virtual Network Gateway**; the name locks to `GatewaySubnet`.

2 Starting address `10.10.255.0`, size **/27**; NAT gateway, NSG and route table *None*. Select **Add**.

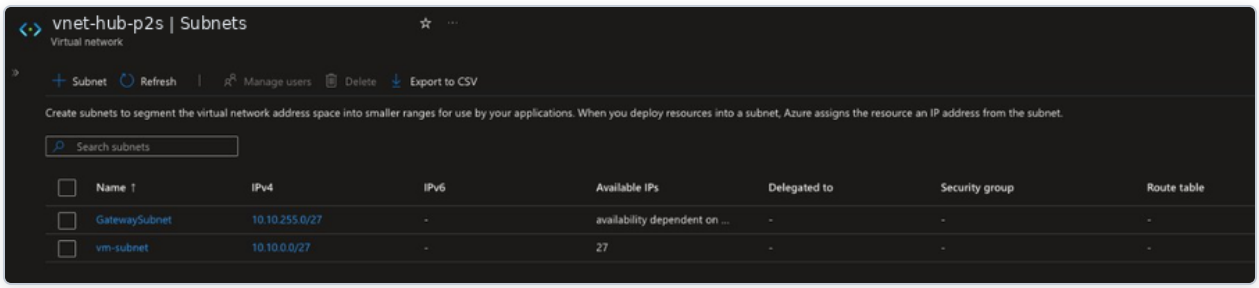


Figure 9. Both subnets exist. "Availability dependent on..." for GatewaySubnet is normal before the gateway is built. (The reference build's workload subnet was resized later; yours should match your plan.)

STEP 3 Create the VPN gateway (30–45 minutes)

1 Search **Virtual network gateways** (**Hybrid connectivity › VPN gateways**) › **+ Create**, and fill in Basics as below.

2 **Review + create** › **Create**. Do not cancel; when done, the gateway and public IP both show *OK*.

Field	Value and reason
Name / Region	`vpngw-hub-p2s`; same region as the VNet, or the VNet will not be listed.
Gateway type / VPN type	VPN / Route-based (required for P2S).
SKU / Generation	VpnGw1AZ / Generation1.
Virtual network	`vnet-hub-p2s`; GatewaySubnet is picked automatically.
Public IP	Create new `pip-vpngw-hub-p2s` (Standard, static, zone-redundant).
Active-active / BGP	Disabled; not needed for P2S alone.

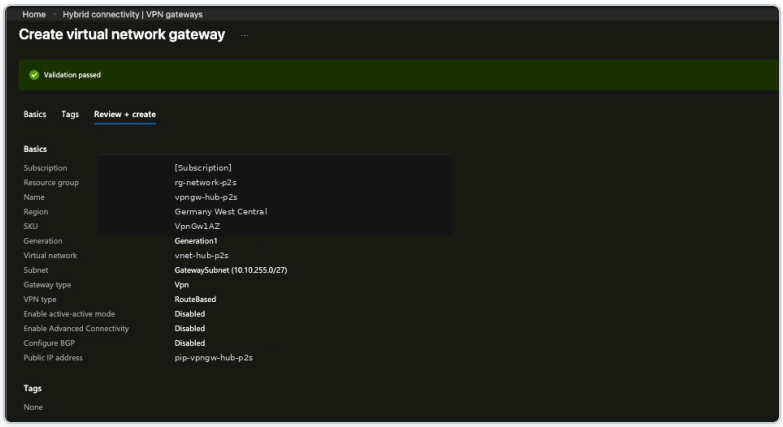


Figure 10. Gateway Review + create showing Generation1, RouteBased and the /27 GatewaySubnet.

STEP 4 Configure Point-to-site settings

1. In the gateway: **Settings › Point-to-site configuration › Configure now**.
2. **Address pool** `172.16.10.0/24`; **Tunnel type** **OpenVPN (SSL)**; **Authentication type** **Microsoft Entra ID** (older portals say *Azure Active Directory*).
3. Enter the Entra fields, using the same tenant ID in both URLs and keeping the trailing slashes:
   - **Tenant** `https://login.microsoftonline.com/<tenant-id>/`
   - **Audience** `c632b3df-fb67-4d84-bdcf-b95ad541b5c8` (or legacy `41b23e61-6c1e-4545-b367-cd054e0ed4b4`)
   - **Issuer** `https://sts.windows.net/<tenant-id>/`
4. **Save**. After about a minute **Download VPN client** becomes active.

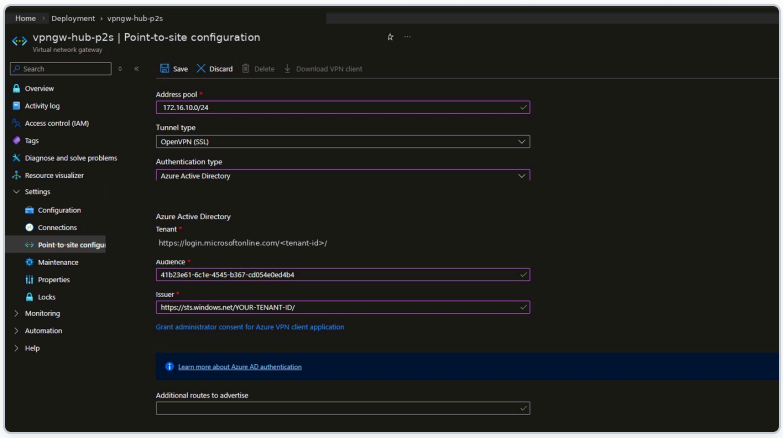


Figure 11. Point-to-site configuration in the reference build, which used the legacy audience; hence the link to grant administrator consent.

WATCH OUT · PLACEHOLDER TEXT

A leftover placeholder such as `YOUR-TENANT-ID` in Tenant or Issuer makes the save fail with a misleading "invalid IP or CIDR notation" error.

STEP 5 Grant admin consent (legacy audience only)

Skip this step with the Microsoft-registered audience. For `41b23e61-…`, consent once per tenant:

1. Select **Grant administrator consent for Azure VPN client application** on the P2S page, or open this URL in a private window:

ADMIN CONSENT URL (LEGACY AUDIENCE, ONE LINE) TEXT

```
https://login.microsoftonline.com/<tenant-id>/oauth2/authorize?client_id=41b23e61-6c1e-4545-b367-
cd054e0ed4b4&response_type=code&redirect_uri=https://portal.azure.com&nonce=1234&prompt=admin_consent
```

2. Sign in as Cloud Application Administrator, tick **Consent on behalf of your organization**, select **Accept**.
3. Verify that **Entra ID › Enterprise applications** now lists *Azure VPN*.

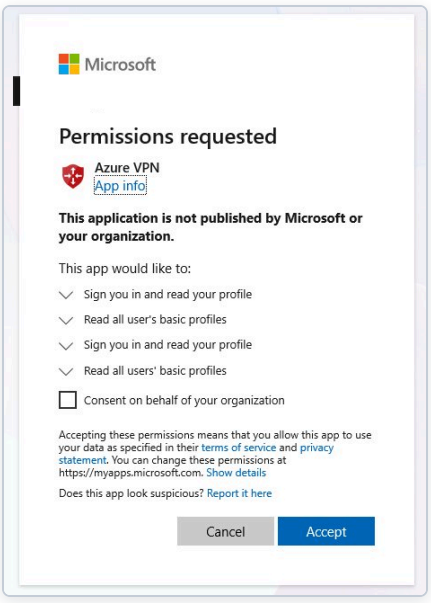


Figure 12. Consent prompt for the legacy app. The "not published by Microsoft" warning is expected for this ID.

STEP 6 Lock down access, then distribute the profile

1. **Restrict who can connect**: in **Entra ID › Enterprise applications** find the VPN app (search by App ID), set **Properties › Assignment required** = **Yes**, and assign `SG-VPN-P2S-Users`.
2. **Require MFA**: **Entra ID › Conditional Access › + New policy** targeting the VPN group and app; grant with *Require multifactor authentication* (optionally a compliant device). Run in *Report-only* first, then *On*.
3. **Allow RDP only from the pool**: NSG `nsg-vm-subnet` with an inbound rule from `172.16.10.0/24` to `10.10.1.0/24`, TCP 3389, Allow, priority 200, associated with `vm-subnet`. Never allow 3389 from *Any* or *Internet*. (The default *AllowVnetInBound* rule already admits the pool via the `VirtualNetwork` tag; the explicit rule documents intent and survives later segmentation.)
4. Select **Download VPN client**, extract the ZIP and take `AzureVPN\azurevpnconfig.xml`. Distribute it via a restricted SharePoint library or Intune; it is not a credential, but should not be public. Any later P2S change means re-downloading and redistributing it.

CHAPTER 4

Methods B, C and D · PowerShell, Azure CLI and Certificates

4.1 Method B · Az PowerShell base template

The template builds Steps 1–4 and 6 (network side) and downloads the profile. The Entra-side tasks (consent, assignment, Conditional Access) are tenant-wide and usually owned by the identity team, so the script reminds rather than performs them. It supports `-WhatIf`, tags everything, and writes a JSON evidence file.

NEW-AZP2SVPN.PS1 · BASE TEMPLATE POWERSHELL

```powershell
<#
.SYNOPSIS  Build an Azure P2S VPN (OpenVPN + Microsoft Entra ID auth) in front of a private workload
subnet.
.NOTES     Requires Az.Accounts, Az.Resources, Az.Network (PowerShell 7+ or Cloud Shell).
           Roles: Network Contributor on the RG (Contributor on the subscription to create the RG).
           Gateway creation takes 30-45 minutes.
#>
[CmdletBinding(SupportsShouldProcess)]
param(
  [Parameter(Mandatory)] [string] $SubscriptionId,
  [Parameter(Mandatory)] [string] $TenantId,                    # Entra tenant GUID
  [Parameter(Mandatory)] [string] $ChangeRef,                   # e.g. CHG0012345
  [string] $ResourceGroup   = 'rg-network-p2s',
  [string] $Location        = 'germanywestcentral',
  [string] $VnetName        = 'vnet-hub-p2s',
  [string] $VnetPrefix      = '10.10.0.0/16',
  [string] $VmSubnetPrefix  = '10.10.1.0/24',
  [string] $GwSubnetPrefix  = '10.10.255.0/27',
  [string] $ClientPool      = '172.16.10.0/24',
  [string] $GatewayName     = 'vpngw-hub-p2s',
  [ValidateSet('VpnGw1AZ','VpnGw2AZ','VpnGw3AZ')] [string] $GatewaySku = 'VpnGw1AZ',
  [ValidateSet('Microsoft','Legacy')]             [string] $Audience   = 'Microsoft'
)
$ErrorActionPreference = 'Stop'
Connect-AzAccount -Tenant $TenantId -Subscription $SubscriptionId | Out-Null
$tags = @{ ChangeRef = $ChangeRef; Owner = (Get-AzContext).Account.Id; Workload = 'P2S-VPN' }
$audienceId = if ($Audience -eq 'Microsoft') { 'c632b3df-fb67-4d84-bdcf-b95ad541b5c8' }
              else { '41b23e61-6c1e-4545-b367-cd054e0ed4b4' }

# ---- Stage 1: resource group ---------------------------------------------
if (-not (Get-AzResourceGroup -Name $ResourceGroup -ErrorAction SilentlyContinue) -and
    $PSCmdlet.ShouldProcess($ResourceGroup, 'New-AzResourceGroup')) {
  New-AzResourceGroup -Name $ResourceGroup -Location $Location -Tag $tags | Out-Null
}

# ---- Stage 2: NSG that only allows RDP from the VPN client pool ----------
$rdp = New-AzNetworkSecurityRuleConfig -Name 'Allow-RDP-from-P2S' -Access Allow -Protocol Tcp `
         -Direction Inbound -Priority 200 -SourceAddressPrefix $ClientPool -SourcePortRange '*' `
         -DestinationAddressPrefix $VmSubnetPrefix -DestinationPortRange 3389
if ($PSCmdlet.ShouldProcess('nsg-vm-subnet', 'New-AzNetworkSecurityGroup')) {
  $nsg = New-AzNetworkSecurityGroup -Name 'nsg-vm-subnet' -ResourceGroupName $ResourceGroup `
           -Location $Location -SecurityRules $rdp -Tag $tags -Force
}
```

```
# ---- Stage 3: VNet with vm-subnet (NSG attached) and GatewaySubnet ----------
if ($PSCmdlet.ShouldProcess($VnetName, 'New-AzVirtualNetwork')) {
  $vmSub = New-AzVirtualNetworkSubnetConfig -Name 'vm-subnet' -AddressPrefix $VmSubnetPrefix -
NetworkSecurityGroup $nsg
  $gwSub = New-AzVirtualNetworkSubnetConfig -Name 'GatewaySubnet' -AddressPrefix $GwSubnetPrefix  # exact
name required
  $vnet  = New-AzVirtualNetwork -Name $VnetName -ResourceGroupName $ResourceGroup -Location $Location `
             -AddressPrefix $VnetPrefix -Subnet $vmSub, $gwSub -Tag $tags -Force
}

# ---- Stage 4: Standard, static, zone-redundant public IP --------------------
if ($PSCmdlet.ShouldProcess("pip-$GatewayName", 'New-AzPublicIpAddress')) {
  $pip = New-AzPublicIpAddress -Name "pip-$GatewayName" -ResourceGroupName $ResourceGroup -Location
$Location `
           -AllocationMethod Static -Sku Standard -Zone 1,2,3 -Tag $tags -Force
}

# ---- Stage 5: the VPN gateway (30-45 min) -----------------------------------
if ($PSCmdlet.ShouldProcess($GatewayName, 'New-AzVirtualNetworkGateway')) {
  $gwSubnet = Get-AzVirtualNetworkSubnetConfig -Name 'GatewaySubnet' -VirtualNetwork $vnet
  $ipConf   = New-AzVirtualNetworkGatewayIpConfig -Name 'gwipconfig1' -SubnetId $gwSubnet.Id -
PublicIpAddressId $pip.Id
  New-AzVirtualNetworkGateway -Name $GatewayName -ResourceGroupName $ResourceGroup -Location $Location `
    -IpConfigurations $ipConf -GatewayType Vpn -VpnType RouteBased -GatewaySku $GatewaySku `
    -VpnGatewayGeneration Generation1 -Tag $tags | Out-Null
}

# ---- Stage 6: Point-to-site settings (OpenVPN + Entra ID) and profile -------
if ($PSCmdlet.ShouldProcess($GatewayName, 'Set-AzVirtualNetworkGateway (P2S)')) {
  $gw = Get-AzVirtualNetworkGateway -Name $GatewayName -ResourceGroupName $ResourceGroup
  Set-AzVirtualNetworkGateway -VirtualNetworkGateway $gw `
    -VpnClientAddressPool $ClientPool -VpnClientProtocol OpenVPN `
    -AadTenantUri "https://login.microsoftonline.com/$TenantId/" `
    -AadAudienceId $audienceId `
    -AadIssuerUri "https://sts.windows.net/$TenantId/" | Out-Null
  # 'EapTls' is a legacy parameter value; the ZIP still contains the Entra profile in the AzureVPN folder
  $cfg = New-AzVpnClientConfiguration -ResourceGroupName $ResourceGroup -Name $GatewayName -
AuthenticationMethod EapTls
  Invoke-WebRequest -Uri $cfg.VpnProfileSASUrl -OutFile ".\$ChangeRef-vpnclient.zip"
}

# ---- Evidence for the change record -----------------------------------------
$gw = Get-AzVirtualNetworkGateway -Name $GatewayName -ResourceGroupName $ResourceGroup -ErrorAction
SilentlyContinue
[pscustomobject]@{
  ChangeRef  = $ChangeRef; Operator = (Get-AzContext).Account.Id; TimeUtc = (Get-Date).ToUniversalTime()
  GatewayId  = $gw.Id; Sku = $gw.Sku.Name; ProvisioningState = $gw.ProvisioningState
  ClientPool = $gw.VpnClientConfiguration.VpnClientAddressPool.AddressPrefixes
  Protocols  = $gw.VpnClientConfiguration.VpnClientProtocols; Audience = $audienceId
} | ConvertTo-Json | Out-File ".\$ChangeRef-p2s-evidence.json"
if ($Audience -eq 'Legacy') { Write-Warning 'Legacy audience: grant admin consent once (Chapter 3, Step
5).' }
Write-Host 'Next: assignment + Conditional Access (Step 6), then distribute AzureVPN\azurevpnconfig.xml.'
-ForegroundColor Green
```

EXAMPLE · RUNNING THE TEMPLATE

INVOCATION POWERSHELL

```
.\New-AzP2SVpn.ps1 -SubscriptionId '<subscription-id>' -TenantId '<tenant-id>' -ChangeRef
'CHG0012345' `
  -Location 'germanywestcentral' -GatewaySku VpnGw1AZ -Audience Microsoft -WhatIf
```

Run with `-WhatIf` to list what would be created, then for real, in Cloud Shell or a session that will not sleep during the 45-minute gateway build.

4.2 Method C · Azure CLI

The same build in Bash, pre-installed in Cloud Shell and natural in GitHub Actions or Azure DevOps. `--no-wait` plus `az network vnet-gateway wait` lets a pipeline do other work while the gateway deploys.

NEW-P2S-VPN.SH BASH

```
RG=rg-network-p2s; LOC=germanywestcentral; VNET=vnet-hub-p2s; GW=vpngw-hub-p2s; CHG=CHG0012345
TENANT=<tenant-id>; AUD=c632b3df-fb67-4d84-bdcf-b95ad541b5c8      # or 41b23e61-6c1e-4545-b367-
cd054e0ed4b4

az group create -n $RG -l $LOC --tags ChangeRef=$CHG
az network nsg create -g $RG -n nsg-vm-subnet --tags ChangeRef=$CHG
az network nsg rule create -g $RG --nsg-name nsg-vm-subnet -n Allow-RDP-from-P2S --priority 200 \
   --direction Inbound --access Allow --protocol Tcp --source-address-prefixes 172.16.10.0/24 \
   --destination-address-prefixes 10.10.1.0/24 --destination-port-ranges 3389
az network vnet create -g $RG -n $VNET --address-prefixes 10.10.0.0/16 \
   --subnet-name vm-subnet --subnet-prefixes 10.10.1.0/24 --network-security-group nsg-vm-subnet --tags
ChangeRef=$CHG
az network vnet subnet create -g $RG --vnet-name $VNET -n GatewaySubnet --address-prefixes 10.10.255.0/27
az network public-ip create -g $RG -n pip-$GW --sku Standard --allocation-method Static --zone 1 2 3

az network vnet-gateway create -g $RG -n $GW --vnet $VNET --public-ip-addresses pip-$GW \
   --gateway-type Vpn --vpn-type RouteBased --sku VpnGw1AZ --vpn-gateway-generation Generation1 \
   --tags ChangeRef=$CHG --no-wait
az network vnet-gateway wait -g $RG -n $GW --created          # 30-45 minutes

az network vnet-gateway update -g $RG -n $GW --address-prefixes 172.16.10.0/24 --client-protocol OpenVPN
\
   --aad-tenant "https://login.microsoftonline.com/$TENANT/" --aad-audience $AUD \
   --aad-issuer "https://sts.windows.net/$TENANT/"

URL=$(az network vnet-gateway vpn-client generate -g $RG -n $GW -o tsv)   # SAS URL to the profile ZIP
curl -sSL "$URL" -o $CHG-vpnclient.zip
```

4.3 Method D · Certificate authentication (alternative)

Certificates prove the *device* rather than an Entra account. Use them for people without Entra accounts, for IKEv2 native clients, or as a break-glass path. They can run alongside

Entra ID on the same gateway. The trade-off: no Conditional Access or MFA, and you own the PKI lifecycle (issuance, renewal, revocation).

1. **Root certificate**: issue one from your enterprise PKI (self-signed only for labs, as below).
2. **Upload the root's public key** (Base-64) under **Point-to-site configuration › Root certificates**, add *Azure certificate* as an authentication type, and choose *IKEv2 and OpenVPN* tunnels.
3. **Client certificates** (with private keys) go on each device, ideally via an Intune PKCS or SCEP profile. Re-download the profile; it now also contains native IKEv2 and OpenVPN settings.
4. **Revoke** a lost device by adding its certificate thumbprint to the revoked list.

LAB CERTIFICATES + UPLOAD (OUTLINE) — POWERSHELL

```powershell
# Self-signed root (lab only - use enterprise PKI in production)
$root = New-SelfSignedCertificate -Type Custom -KeySpec Signature -Subject 'CN=P2SRootCert' `
          -KeyExportPolicy Exportable -HashAlgorithm sha256 -KeyLength 2048 `
          -CertStoreLocation 'Cert:\CurrentUser\My' -KeyUsageProperty Sign -KeyUsage CertSign
# Client certificate signed by the root, with the Client Authentication EKU
New-SelfSignedCertificate -Type Custom -DnsName 'P2SClient01' -KeySpec Signature -Subject
'CN=P2SClient01' `
  -KeyExportPolicy Exportable -HashAlgorithm sha256 -KeyLength 2048 -CertStoreLocation
'Cert:\CurrentUser\My' `
  -Signer $root -TextExtension @('2.5.29.37={text}1.3.6.1.5.5.7.3.2')
# Upload only the PUBLIC part of the root
Add-AzVpnClientRootCertificate -ResourceGroupName 'rg-network-p2s' -VirtualNetworkGatewayName 'vpngw-hub-
p2s' `
  -VpnClientRootCertificateName 'P2SRootCert' -PublicCertData ([Convert]::ToBase64String($root.RawData))
# Revoke a client later
Add-AzVpnClientRevokedCertificate -ResourceGroupName 'rg-network-p2s' -VirtualNetworkGatewayName 'vpngw-
hub-p2s' `
  -VpnClientRevokedCertificateName 'P2SClient01' -Thumbprint '<client-cert-thumbprint>'
```

A third option, RADIUS, points the gateway at Windows NPS (reachable through the VNet) to validate Active Directory credentials, with MFA via the NPS extension.

CHAPTER 5

Method E · End-User Procedure, Verification and Monitoring

5.1 Connecting as an end user

Each user does this once per device; afterwards connecting takes two clicks. The help desk can use this chapter as its runbook.

STEP 7 Install the client and import the profile

1. Windows: `Microsoft Store` › search *Azure VPN Client* › confirm the publisher is **Microsoft Corporation** › `Get`. macOS uses the Mac App Store; Linux uses Microsoft's package repository; managed PCs can get it from Intune.
2. Open the client, select `+` (bottom-left) › `Import`, choose `azurevpnconfig.xml`, then `Save`. Intune can deploy the profile instead, keeping it current automatically.

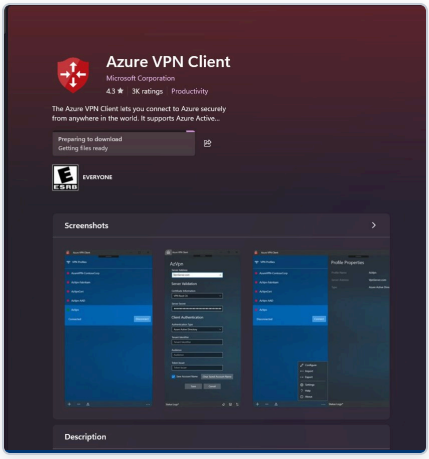


Figure 13. The Azure VPN Client in the Microsoft Store. Always check the publisher: look-alike VPN apps are a phishing lure.

STEP 8 Connect and open Remote Desktop

1. Select the profile › `Connect`, sign in with the work account and approve MFA. Wait for **Connected**; the VPN IP should be `172.16.10.x` and the routes should include `10.10.0.0/16`.

2. Get the VM's private IP (**Virtual machines › <VM> › Overview**), e.g. `10.10.1.4`, and run `mstsc /v:10.10.1.4` (or Windows App / another RDP client).
3. When finished, close RDP and select **Disconnect** in the client.

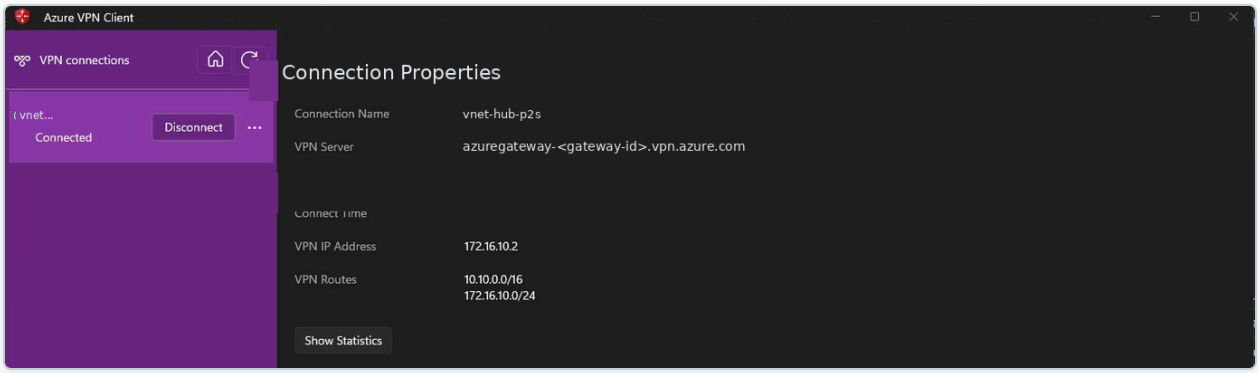


Figure 14. A successful connection: an address from the pool and a route to the VNet, so traffic for Azure uses the tunnel.

QUICK CLIENT-SIDE CHECKS — POWERSHELL

```
Get-NetIPAddress -AddressFamily IPv4 | Where-Object IPAddress -like '172.16.10.*'   # tunnel address present
Test-NetConnection 10.10.1.4 -Port 3389                                            # TcpTestSucceeded : True
```

5.2 Verification checklist

Azure and Entra side

- ☐ Gateway *Succeeded*; SKU and generation as planned
- ☐ P2S: pool, OpenVPN, Entra ID, correct audience
- ☐ VPN app: *Assignment required* = *Yes*, group assigned
- ☐ CA policy *On* and visible in sign-in logs
- ☐ NSG allows 3389 only from the pool; no VM public IPs

User side

- ☐ Client *Connected* with a 172.16.10.x address
- ☐ Route to 10.10.0.0/16 present
- ☐ `Test-NetConnection` to 3389 succeeds
- ☐ RDP opens to the private IP
- ☐ An unassigned test user is refused

5.3 Sessions and logs

VPN gateway › Point-to-site sessions lists connected users with their tunnel IPs and lets you disconnect one. Send *P2SDiagnosticLog*, *GatewayDiagnosticLog* and *TunnelDiagnosticLog*

to Log Analytics via **Diagnostic settings**, and pair them with Entra sign-in logs, which record each VPN sign-in, the MFA result and the CA decision.

SESSIONS AND FORCED DISCONNECT — POWERSHELL

```powershell
Get-AzVirtualNetworkGatewayVpnClientConnectionHealth -VirtualNetworkGatewayName 'vpngw-hub-p2s' -
ResourceGroupName 'rg-network-p2s' |
  Select-Object UserName, VpnConnectionId, PrivateIpAddress, VpnConnectionTime
Disconnect-AzVirtualNetworkGatewayVpnConnection -VirtualNetworkGatewayName 'vpngw-hub-p2s' -
ResourceGroupName 'rg-network-p2s' `
  -VpnConnectionId '<connection-id>'      # e.g. after disabling a leaver's account
```

RECENT P2S EVENTS (LOG ANALYTICS) — KQL

```kql
AzureDiagnostics
| where ResourceType == "VIRTUALNETWORKGATEWAYS" and Category == "P2SDiagnosticLog"
| where TimeGenerated > ago(24h)
| project TimeGenerated, Message | order by TimeGenerated desc
```

5.4 Control alignment (NIST SP 800-53 Rev. 5)

Control	How this design supports it
AC-17	Remote Access: one authenticated, encrypted, logged path for remote administration.
AC-2	Account Management: access bound to individual Entra accounts and group assignment; disabling an account blocks new connections.
IA-2 (1)(2)	MFA enforced by Conditional Access for every VPN sign-in.
IA-3	Device Identification: met only with a CA device-compliance requirement or certificate auth. The XML profile is not a device credential.
SC-7 / CM-7	Boundary Protection / Least Functionality: no VM public IPs, the gateway is the only internet-facing component, and only required ports are open from the pool.
SC-8	Transmission Confidentiality: OpenVPN uses TLS over TCP 443.
AU-2 / AU-12	Logging: gateway diagnostics, Activity Log and Entra sign-in logs.

CHAPTER 6

Rollback, Cost and Troubleshooting

6.1 Rollback and cost

- **Wrong P2S settings or a stale profile**: edit, save, re-download and redistribute the profile. If users are refused at busy times, enlarge the pool or resize the SKU.
- **Abandon the build**: delete the gateway first (10+ minutes), then the public IP, VNet and NSG, or remove everything tagged with the `ChangeRef`. Delete the legacy *Azure VPN* enterprise app to revoke consent.
- **Cost**: the gateway bills every hour from creation to deletion and cannot be stopped; the Standard public IP and outbound data also bill. For a few admins needing console access only, compare Bastion.

6.2 Troubleshooting

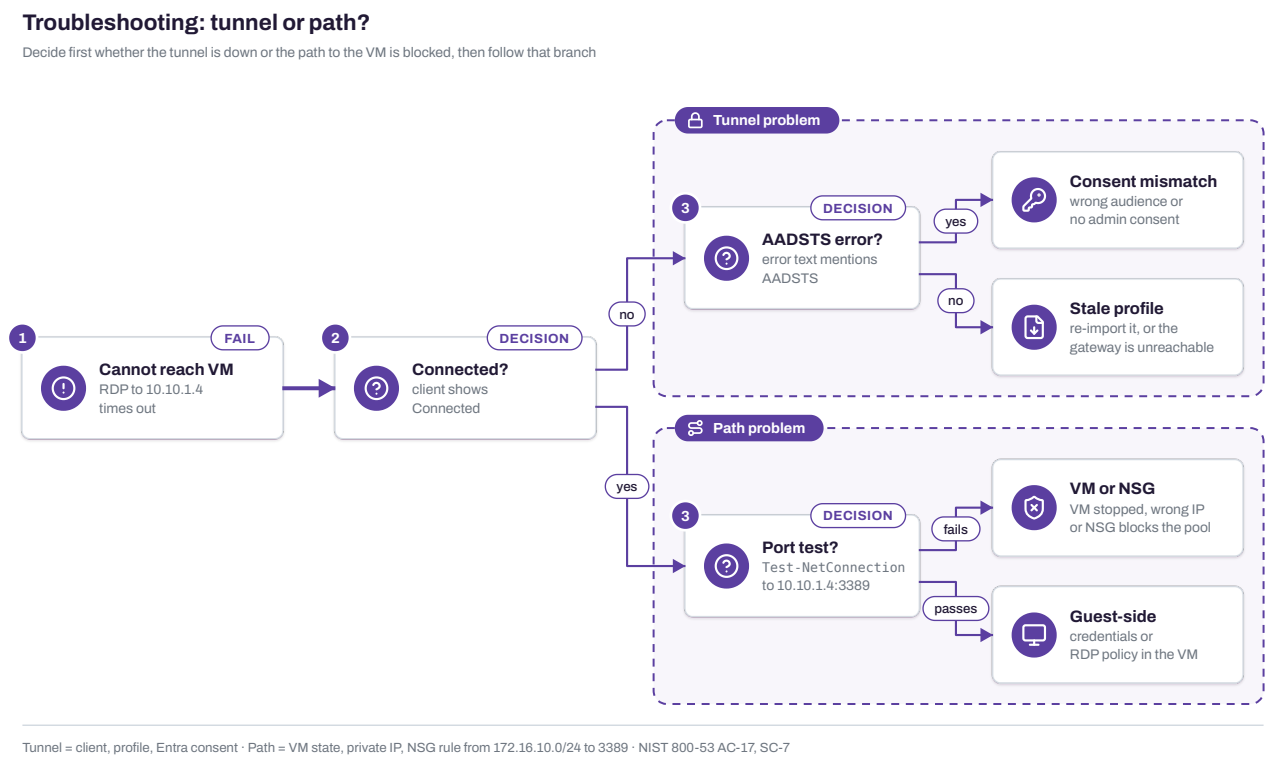


Figure 15. Splitting the problem into tunnel or path halves the search before you open a single log.

Symptom	Likely cause and fix
`AADSTS90008` **"has not consented"**	Legacy audience without tenant consent. Grant consent (Step 5) or switch to the Microsoft-registered audience and redistribute profiles.
`AADSTS50105` **user not assigned**	*Assignment required* is on and the user is not in the assigned group.
Conditional Access blocks sign-in	Read the user's sign-in log: missing MFA, non-compliant device or blocked location.
Save fails: "invalid IP or CIDR notation"	Placeholder, typo or mismatched tenant ID in Tenant/Issuer; both need the same GUID and a trailing slash.
Connected, RDP times out	VM stopped, wrong IP, NSG missing the pool rule, home LAN overlaps the VNet, or guest firewall blocks 3389.
Connected, names do not resolve	No custom DNS on the VNet; use IPs or deploy DNS Private Resolver and reconnect.

CHAPTER 7

Summary, Glossary and Review

7.1 Chapter summary

- P2S VPN gives individual devices an encrypted route into a private VNet, so VMs never need public IPs.
- Plan three non-overlapping ranges (VNet, GatewaySubnet, client pool), use a route-based AZ SKU, and allow 30–45 minutes to deploy.
- OpenVPN + Entra ID brings MFA, Conditional Access and instant revocation. Prefer the Microsoft-registered audience; the legacy one needs one-time admin consent.
- Restrict with app assignment, CA and narrow NSG rules; keep gateway and sign-in logs for audit.

7.2 Glossary

P2S / S2S VPN	Point-to-Site: one device to a gateway. Site-to-Site: an on-premises network to a gateway.
GatewaySubnet	Mandatory, dedicated subnet that hosts the gateway's managed instances.
OpenVPN	TLS-based tunnel protocol on TCP 443; required for Entra ID authentication.
Audience	The application ID an access token must be issued for.
Conditional Access	Entra policy engine requiring MFA, compliant devices or other conditions before issuing tokens.

Check your understanding

1. A contractor needs browser-only access to one jump server for a week. P2S or Bastion, and why?
2. Why must the client pool not overlap the VNet, and what happens if a user's home LAN uses the VNet's range?
3. Which tunnel protocol is required for Entra ID authentication, and on which port does it run?
4. Explain the difference between the `c632b3df-…` and `41b23e61-…` audiences and when admin consent is needed.
5. Which Azure RBAC role and which Entra role are needed to build the gateway and to grant consent, and at what scope?